Polynomial End Behavior & Zeros

Section 1: Leading Term Test and End Behavior

1 Determine the end behavior of the polynomial function $f(x) = -3x^5 + 7x^3 - 2x + 9$.

a. As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$

b. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$

c. As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$

d. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to \infty$

2 If a polynomial $p(x)$ satisfies $\lim_{x \to \infty} p(x) = -\infty$ and $\lim_{x \to -\infty} p(x) = -\infty$, which of the following could be the leading term of $p(x)$?

a. $-4x^6$

b. $2x^6$

c. $-5x^7$

d. $3x^5$

3 Consider $g(x) = (2 - x)(x + 3)^2(2x - 1)$. What is the leading coefficient of $g(x)$?

a. -2

b. 2

c. -1

d. 4

Section 2: Multiplicity of Zeros and Graph Geometry

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4 Let $f(x) = (x-4)^3(x+2)^2(x-1)$. At which zero does the graph of f touch the x-axis and turn around without crossing?

a. $x = -2$

b. $x = 4$

c. $x = 1$

d. $x = 2$

5 A polynomial $p(x)$ of degree 4 has a zero at $x = 3$ of multiplicity 2, and no other real zeros. If $p(0) = 18$, which equation represents $p(x)$ if its remaining zeros are non-real complex conjugates $\pm i$?

a. $p(x) = 2(x-3)^2(x^2+1)$

b. $p(x) = (x-3)^2(x^2+1)$

c. $p(x) = 2(x-3)^2(x^2-1)$

d. $p(x) = \frac{1}{2}(x-3)^2(x^2+1)$

6 At $x = c$, $f(x) = (x-c)^3 g(x)$ where $g(c) \neq 0$. What is the behavior of the graph of f at $(c, 0)$?

a. The graph crosses the x-axis with an inflection/flattening point (horizontal tangent)

b. The graph touches the x-axis and turns around symmetrically

c. The graph has a vertical asymptote at $x = c$

d. The graph crosses the x-axis linearly without flattening

Section 3: Remainder & Factor Theorems

7 When $f(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x-3)$, the remainder is 7. Find the value of k.

a. -4

b. 4

c. -2

d. 5

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8 Which of the following is a factor of $P(x) = x^3 - 6x^2 + 11x - 6$?

a. $x - 2$

b. $x + 2$

c. $x - 4$

d. $x + 1$

9 A polynomial $q(x)$ satisfies $q(-4) = 0$, $q(1) = 0$, and $q(5) = 12$. Which statement must be true?

a. $(x + 4)$ and $(x - 1)$ are factors of $q(x)$

b. $(x - 4)$ and $(x + 1)$ are factors of $q(x)$

c. The remainder when $q(x)$ is divided by $(x + 5)$ is 12

d. $q(x)$ must be a quadratic polynomial

Section 4: Rational Root Theorem & Complex Conjugates

10 According to the Rational Root Theorem, which of the following is NOT a possible rational zero of $f(x) = 3x^3 - 2x^2 + 5x - 4$?

a. $\frac{3}{2}$

b. $\frac{4}{3}$

c. ± 1

d. ± 2

11 A polynomial with real coefficients has degree 3. Two of its zeros are 4 and $2 - 3i$. What is the third zero?

a. $2 + 3i$

b. $-2 + 3i$

c. -4

d. $-2 - 3i$

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12 What is the total number of complex zeros (counting multiplicity) of $P(x) = (x^2 + 4)^2(x - 5)^3$?

a. 7

b. 5

c. 4

d. 3

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